

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -3

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Recap of Previous Lecture



doubt ?

Topic

Non-Contiguous Memory
Management Technique

Topic

Paging

Topic

Address Translation



Topics to be Covered



Topic

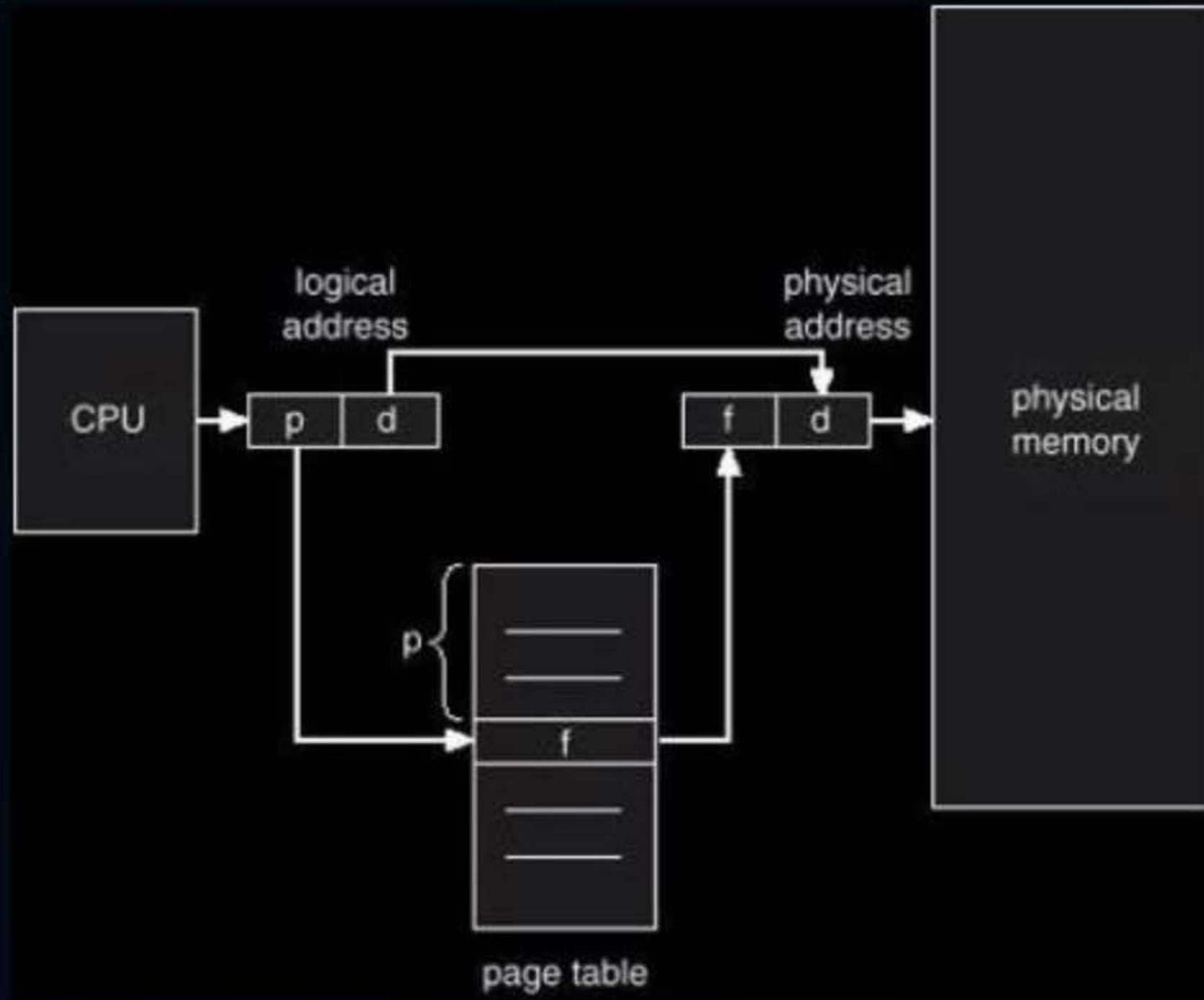
Performance of Paging

Topic

TLB



Topic : Paging



#Q. Consider a paged memory system where the process size is 16MB and main memory size is 4GB. The page size is 2KB.

1. Number of pages in process? 2^{13}
2. Number of frames in main memory? 2^{21}
3. Number of bits for page number? 13
4. Number of bits for frame number? 21
5. Number of entries in page table? 2^{13}
6. Page table size? $2^{13} * 21 \text{ bits}$

#Q. Consider a paged memory system where the process size is 128MB and main memory size is 2GB. The page size is 1KB.

1. Number of pages in process? 2^{17}
2. Number of frames in main memory? 2^{21}
3. Number of bits for page number? 17
4. Number of bits for frame number? 21
5. Number of entries in page table? 2^{17}
6. Page table size? $2^{17} * 21 \text{ bits}$

#Q. Consider a paged memory system where the logical address is 25 bits and physical address is 33 bits. The page size is 4KB.

1. Number of pages in process?
2. Number of frames in main memory?
3. Number of bits for page number?
4. Number of bits for frame number?
5. Number of entries in page table?
6. Page table size?



- #Q. A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the logical address supported by the system is _____ bits?

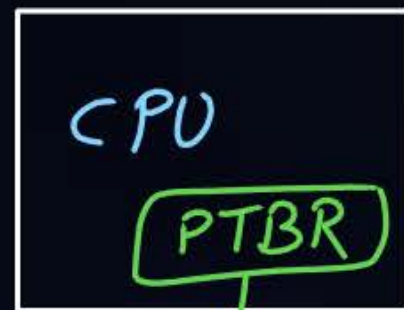




Topic : Paging

Where the Page table Stored?

in main memory



Page table Base Register
(stores starting address
of P.T.)

L.A.

mm



starting add.
of P.T.



Topic : Paging



Performance of Paging

$$\text{Effective mem. access time} = \frac{2 * t_{mm}}{}$$

$t_{mm} \Rightarrow$ mm access time

→ one for P.T.
one for content

if process and its page table are very-very small
↓
keep the P.T. in CPU Registers

$$E.M.A.T. = t_{mm}$$

(P.T. access time from registers
is negligible)



Topic : Paging

TLB (Translation Lookaside Buffer)

used to store most frequent and most used P.T. entries.

CPU generates L.A.

↓
search in TLB

Hit

P.A.

↓
Access mm for
Content

Miss

Access mm for P.T.

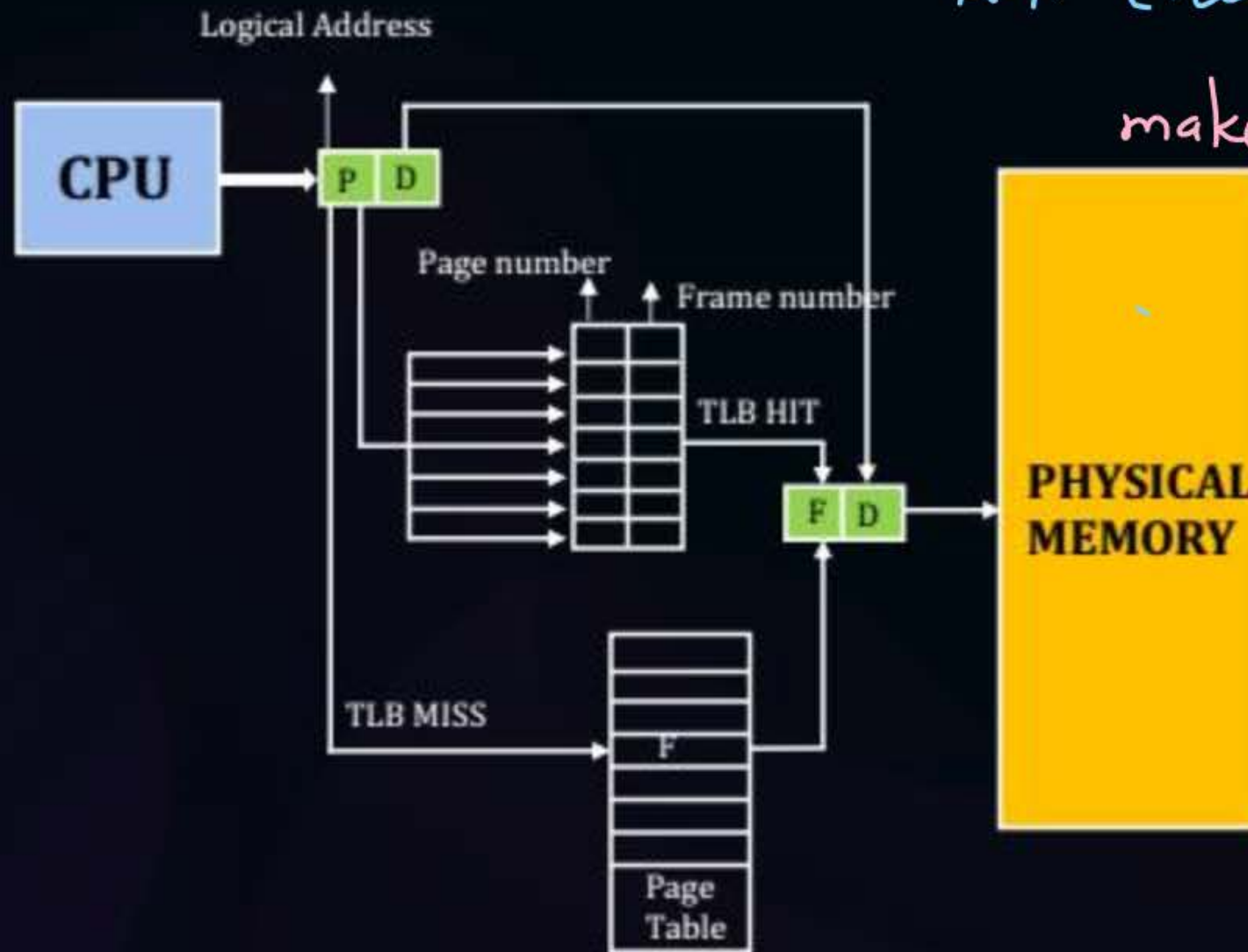
↓
P.A.

↓
Access mm for content



Topic : Paging

TLB (Translation Lookaside Buffer)



when Process Switch in CPU

TLB stores only
P.T. entries

make all
TLB
entries
invalid
(TLB
empty)

TLB stores pid
and TLB entry

keep TLB entries
for future use
in TLB only.

$$\text{E.M.A.T using TLB} = H (t_{\text{TLB}} + t_{\text{mm}}) + (1-H) (t_{\text{TLB}} + t_{\text{mm}} + t_{\text{mm}})$$

or

$$= t_{\text{TLB}} + t_{\text{mm}} + (1-H) t_{\text{mm}}$$

$H = \text{TLB hit rate}$

$t_{\text{TLB}} = \text{TLB access time}$

$t_{\text{mm}} = \text{mm} - || -$

Ques) $H = 90\%$

$$t_{TLB} = 10 \text{ ns}$$

$$t_{mm} = 200 \text{ ns}$$

$$E_{MAT} = \underline{230} \text{ ns?}$$

$$\begin{aligned} E_{MAT} &= 10 + 200 + 0.1 * 200 \\ &= 230 \text{ ns} \end{aligned}$$

$$\begin{aligned} E_{MAT} \text{ w/o TLB} &= 2 * 200 \\ &= 400 \text{ ns} \end{aligned}$$



Topic : Paging



How TLB Stores Entries?

→ TLB mapping $\Rightarrow$ helps in storing P.T. entries at specific location in TLB and also in searching in TLB.



Topic : Paging

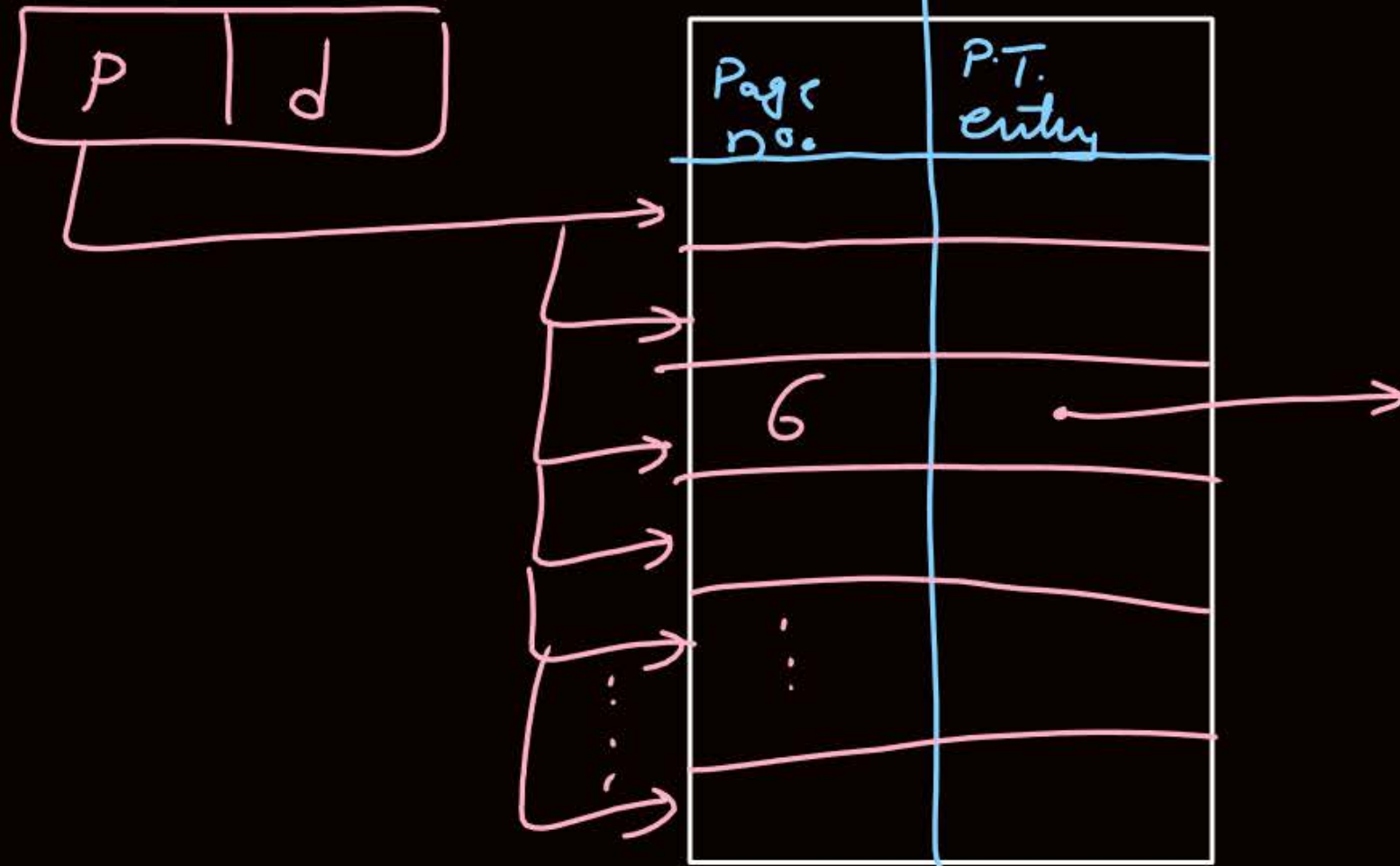


TLB Mapping

1. Fully Associative
2. Direct
3. Set-Associative

fully associative :-

Implemented using associative mem. (content addressable memory)





Topic : Paging



TLB Mapping: Direct

L.A.

P		d
Tag	TLB entry no.	d

if TLB can store no. of entries	TLB entry no.
256	8 bits
128	7 bits
n	$\log_2 n$ bits

ex:-

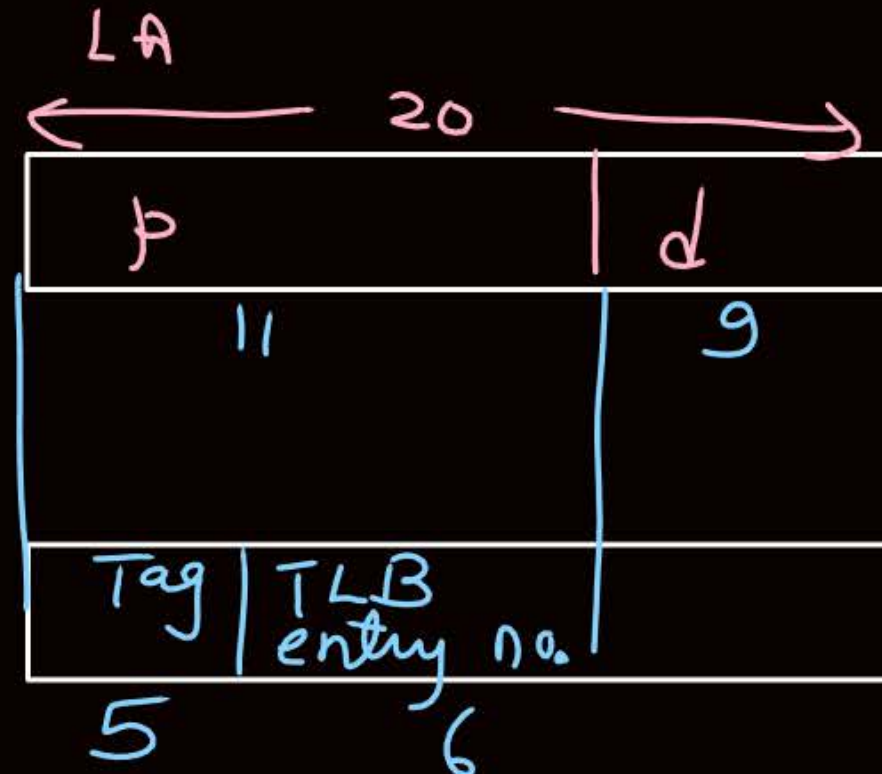
L.A. = 20 bits

Page size = 512 bytes = 2^9 B $\Rightarrow d = 9$ bits

no. of entries in TLB = 64 = 2^6

Direct mapping

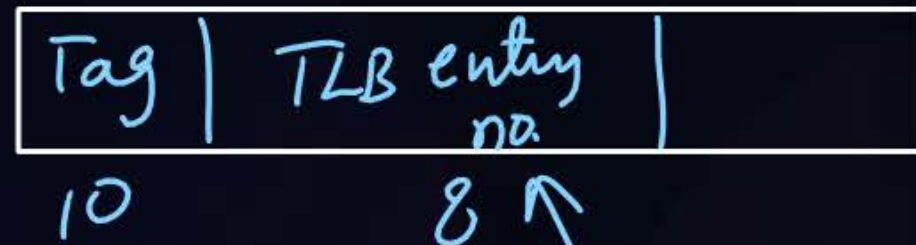
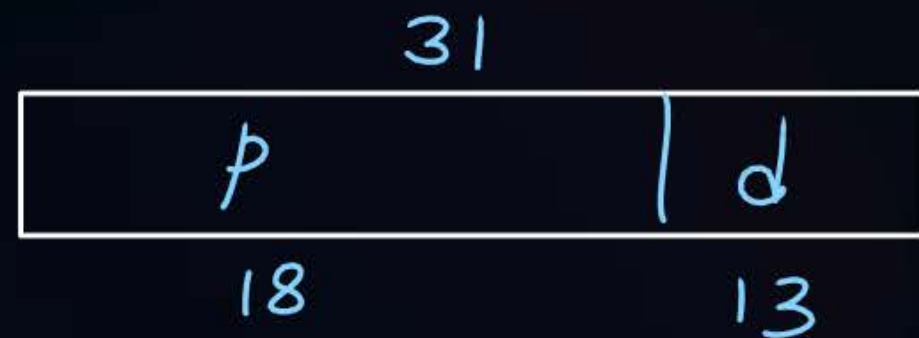
Tag = 5 bits





Topic : Question

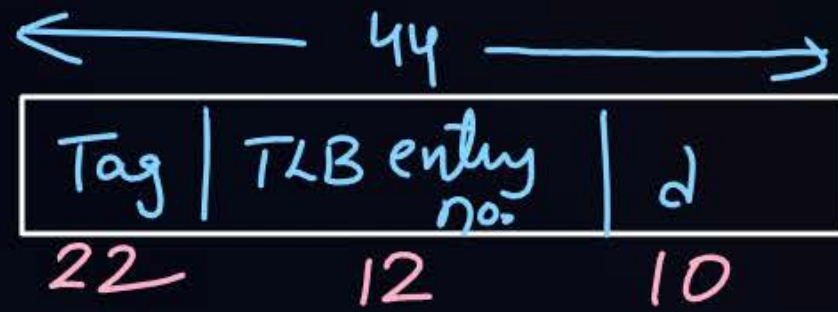
#Q. A computer system implements a 31-bit virtual address, page size of 8 kilobytes, and a 256-entry translation look-aside buffer (TLB) organized as direct mapped. The minimum length of the TLB tag in bits is 10?





Topic : Question

#Q. A computer system implements a 44- bit virtual address, page size of 1 kilobytes, and a 16KB look-aside buffer (TLB) organized as direct mapped. Each page table entry is of 4bytes. The minimum length of the TLB tag in bits is 22?



$$\begin{aligned}\text{no. of entries in TLB} &= \frac{16 \text{ KB}}{4 \text{ B}} \\ &= 4 \text{ K} \\ &= 2^{12}\end{aligned}$$



Topic : Paging



TLB Mapping: Set Associative

L.A.

P		d
Tag	TLB set no.	

$$\text{no. of sets in TLB} = \frac{\text{no. of entries in TLB}}{\text{associativity}}$$

$$\text{bits in TLB set no.} = \log_2 (\text{no. of sets in TLB})$$



Topic : Question

#Q. A Computer system implements a 36-bit virtual address, page size of 16 Kbytes and a 256 – entry translation look-aside buffer (TLB) organized into 64 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is 16.

L.A. $\xleftarrow{36}$

p		d
		14
Tag	TLB set no.	d
16	6	14

$$\begin{aligned}\text{no. of sets in TLB} &= \frac{256}{4} \\ &= 64\end{aligned}$$

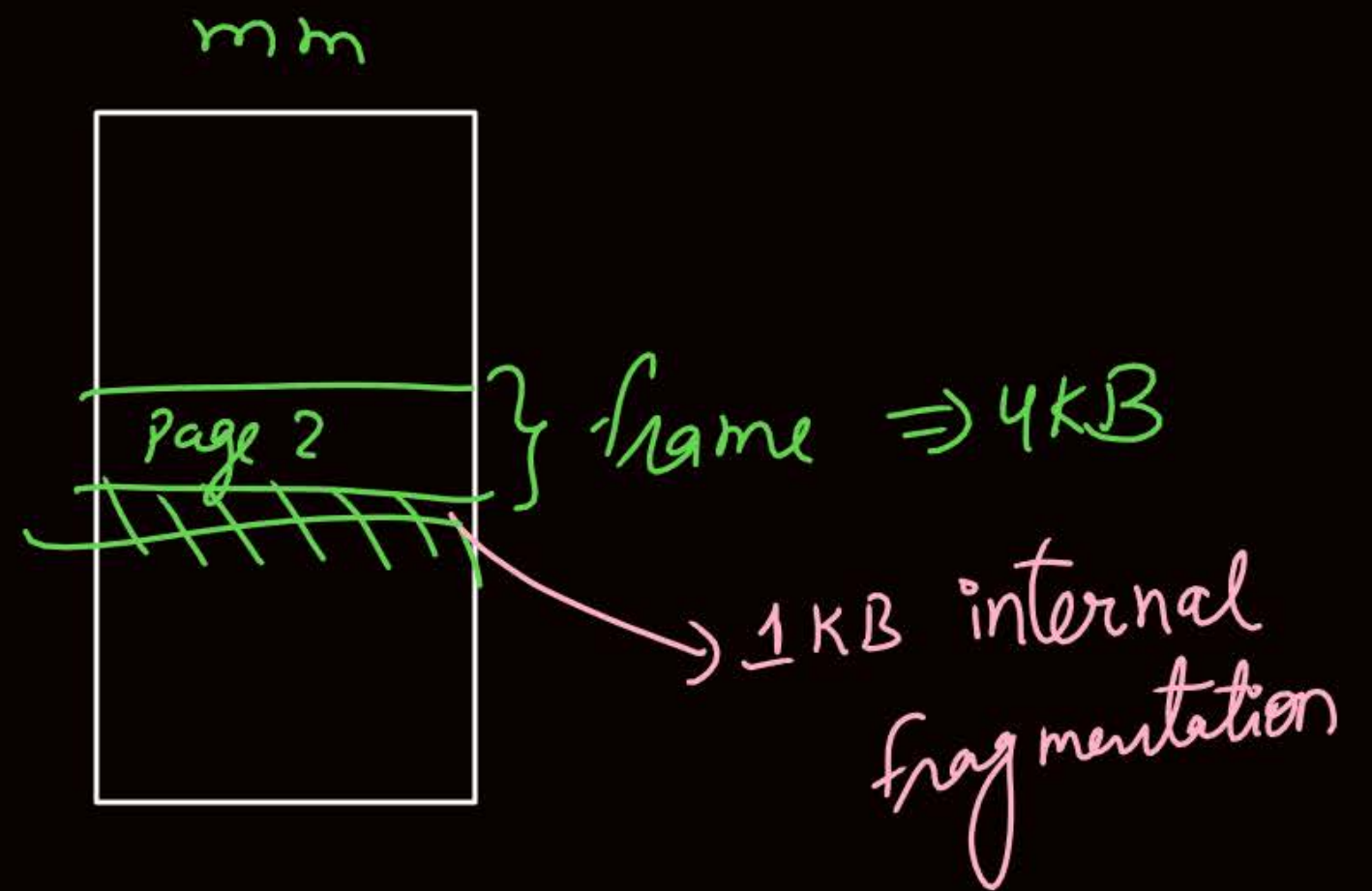
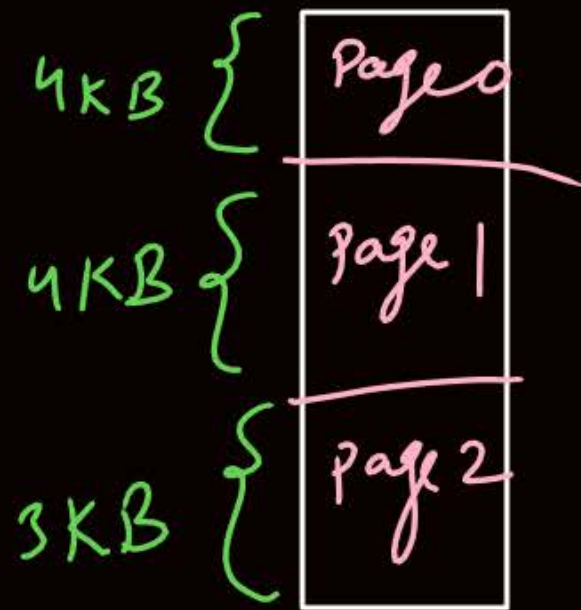
$$2^{14}$$

⇒ Paging suffers from internal fragmentation.

ex:-

Process size = 11 KB

Page size = 4 KB

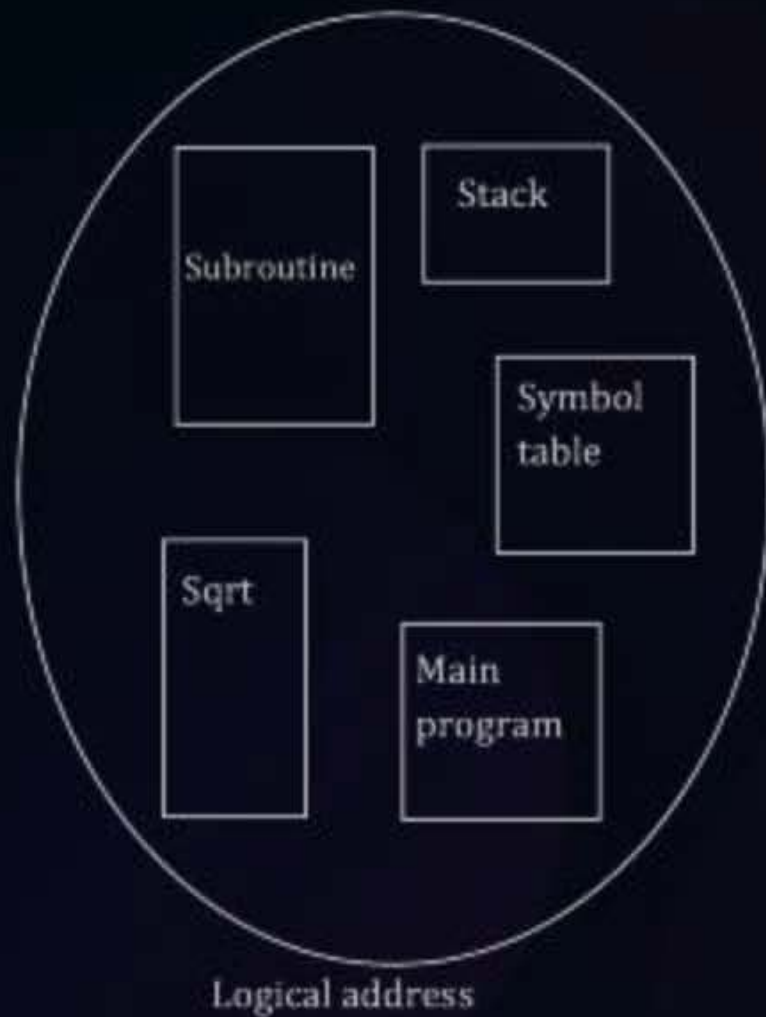




Topic : Segmentation

- Divide Process in logically related partitions (Segments)
- Segments are scattered in physical memory

→ of variable sizes

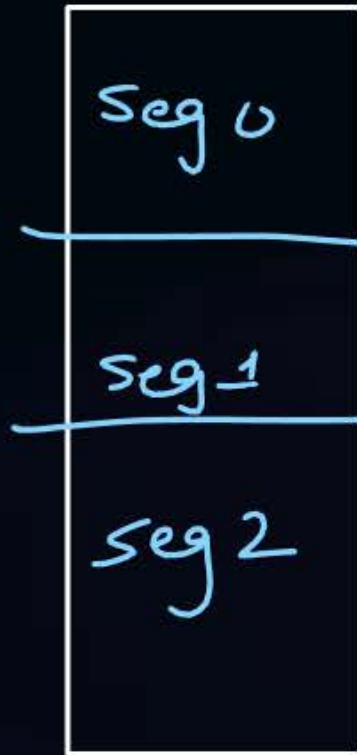




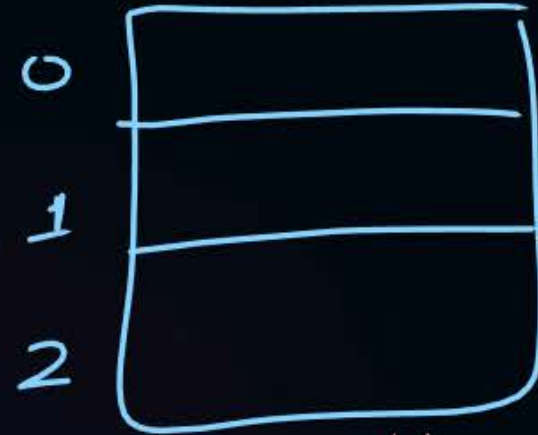
Topic : Segmentation



Process



segment table



Base add
of each segment

mem





Topic : Segmentation

Process
seg 00
seg 01
seg 10
seg 11

	Segment Table	
00	4000	1000
01	3000	500
10	6000	700
11	1000	2000
	Base	Limit

Physical
Add. → Physical Memory



$$P.A. = Base + d$$

$$L.A. \Rightarrow S = 0 \\ d = 1200$$

$$P.A. = \begin{array}{r} 4000 \\ + 1200 \\ \hline 5200 \end{array}$$

↓
invalid add.



Topic : Segmentation

Segment Table

00	4000	1000
01	9000	500
10	6000	700
11	1000	2000
	Base	Limit

Segment number	Offset	Physical Address
1	450	9450
2	752	fault
3	1539	2539
0	996	4996



Topic : Segmentation

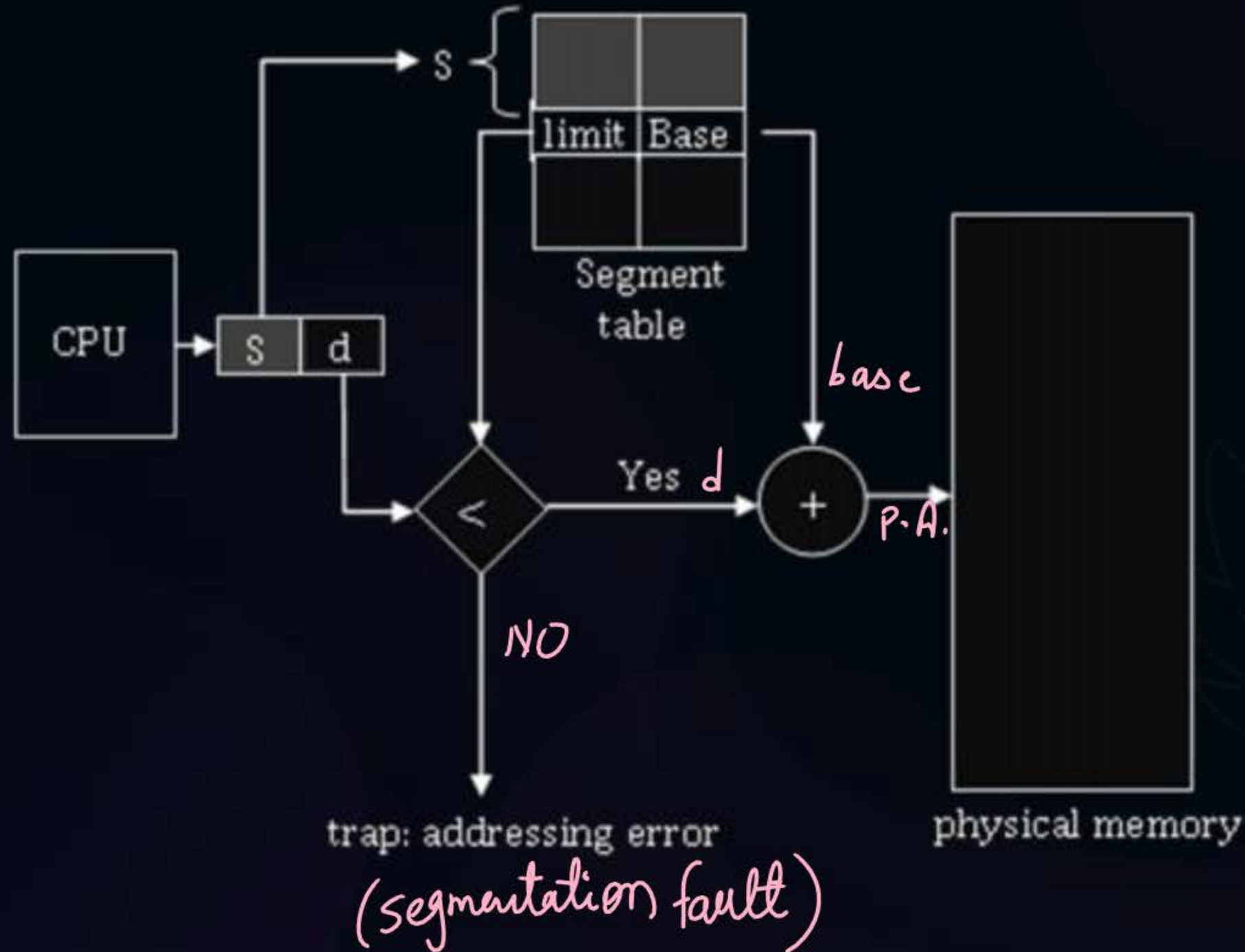


- Size of segment can vary, so along with base, keep limit information also in segment table
- Limit defines max number of words within the segment
or bytes $\Rightarrow$ segment size

bits in offset $\Rightarrow$ max seg. size



Topic : Segmentation



[NAT]



#Q. Maximum segment size = $16\text{KB} = 2^{14}\text{B} \Rightarrow d = 14 \text{ bits}$
Number of segments in process = 2^{10}
Logical address = 24 bits ??





Topic : Segmentation



- Segmentation suffers from external fragmentation



b/w segments the holes are created and one complete segment can not be stored in any of the holes.



2 mins Summary

Topic

Performance of Paging

Topic

TLB



Happy Learning

THANK - YOU

